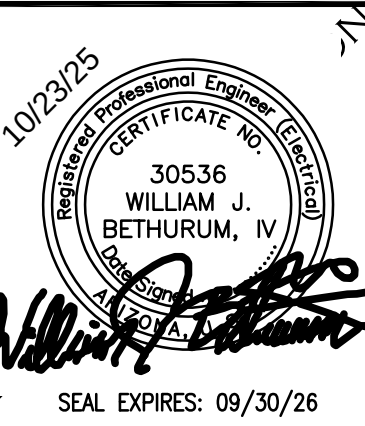


PROJECT ALPHA
1F0 - MAIN MANUFACTURING BUILDING
EAST PECOS ROAD
QUEEN CREEK, ARIZONA 85140

DATA CENTER ONE LINE DIAGRAM

DATE	REMARKS
10/23/24	PRELIMINARY FOR COORDINATION
10/19/2024	DC-074 1F0 CD PROGRESS SET (MEP ONLY)
11/01/2024	DC-074 1F0 FULL CD PACKAGE
02/17/2025	1F0 DC-074 REVISIONS
03/11/2025	1F0 CD REVISIONS
10/23/2025	1F0 BULLETIN #1

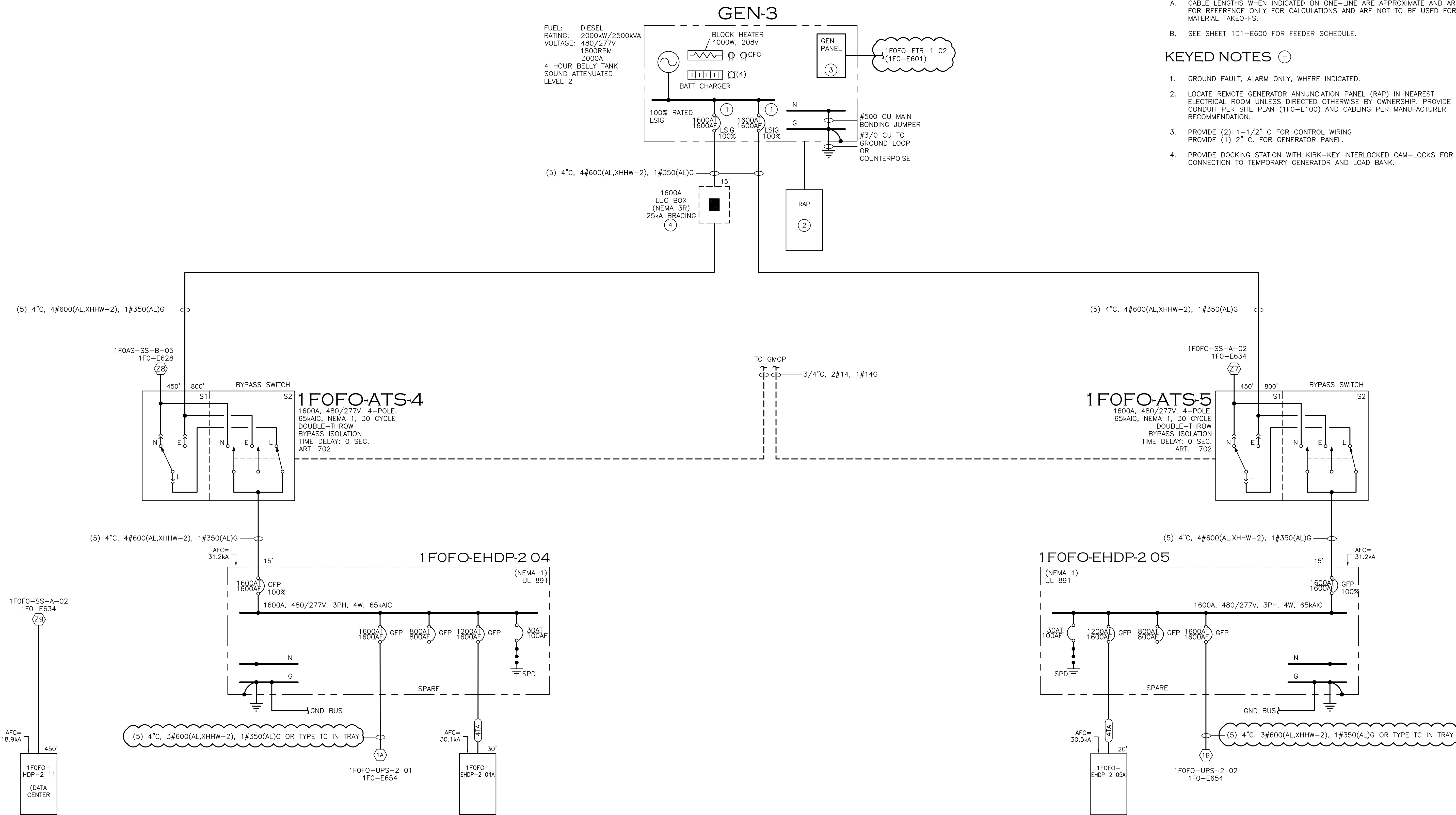
P&P/M:	WB
DRAWN BY:	ETS/AOIA/JG
JOB NO.:	PHX21-5038-00

GENERAL NOTES

- CABLE LENGTHS WHEN INDICATED ON ONE-LINE ARE APPROXIMATE AND ARE FOR REFERENCE ONLY FOR CALCULATIONS AND ARE NOT TO BE USED FOR MATERIAL TAKEOFFS.
- SEE SHEET 101-E600 FOR FEEDER SCHEDULE.

KEYED NOTES

- GROUND FAULT, ALARM ONLY, WHERE INDICATED.
- LOCATE REMOTE GENERATOR ANNUNCIATION PANEL (RAP) IN NEAREST ELECTRICAL ROOM UNLESS DIRECTED OTHERWISE BY OWNERSHIP. PROVIDE CONDUIT PER SITE PLAN (1F0-E100) AND CABLING PER MANUFACTURER RECOMMENDATION.
- PROVIDE (2) 1-1/2" C FOR CONTROL WIRING.
PROVIDE (1) 2" C. FOR GENERATOR PANEL.
- PROVIDE DOCKING STATION WITH KIRK-KEY INTERLOCKED CAM-LOCKS FOR CONNECTION TO TEMPORARY GENERATOR AND LOAD BANK.



ONE LINE DIAGRAM - EMERGENCY POWER DISTRIBUTION
NO SCALE

	RATINGS		CONN.		OPER.		%	
FEEDER ID	VOLTS	AMPS	KVA	DIVERSITY	KVA	AMPS	LOADING	NOTES
M221_M321 ASSEMBLY B	12470	600	5315		4784	221	37%	

	RATINGS		CONN.		OPER.		%	
Bus ID	VOLTS	AMPS	KVA	DIVERSITY	KVA	AMPS	LOADING	NOTES
1FOAS-SS-B-05	480	5000	34.30		3087	3713	74%	
1FOAS-HDP-2 07	480	1200	859	90%	773	930	78%	
1FOAS-HDP-2 08	480	1200	821	90%	739	889	74%	
1FOAS-HDP-2 09	480	1200	839	90%	755	908	76%	
SPARE	480	1600	0	90%	0	0	0%	
1FOAS-HDP-2 03	480	600	169	90%	153	183	31%	
1FOAS-ELP-2 01	480	300	58	90%	52	63	21%	
SPARE	480	1600	0	90%	0	0	0%	
1FOFO-ATS-4 (DATA CENTER)	480	1600	684	90%	615	740	46%	

	RATINGS		CONN.		OPER.		%	
FEEDER ID	VOLTS	AMPS	KVA	DIVERSITY	KVA	AMPS	LOADING	NOTES
M213M313 FORMATION A	12470	800	13797		12418	575	72%	

	RATINGS		CONN.		OPER.		%	
Bus ID	VOLTS	AMPS	KVA	DIVERSITY	KVA	AMPS	LOADING	NOTES
1FOFO-SS-A-02	480	5000	2492		2242	2697	54%	
1FOFO-HDP-2 05	480	800	506	90%	456	548	60%	
1FOFO-HDP-2 06	480	800	397	90%	357	430	54%	
1FOFO-ATS-5 (DATA CENTER)	480	1600	748	90%	673	810	51%	
SPARE	480	1600	0	90%	0	0	0%	
1FOFO-HDP-2 01	480	1200	554	90%	498	599	50%	
1FOFO-HDP-2 11 (DATA CENTER)	480	800	287	90%	258	310	39%	
1FOFO-HDP-2 01	480	1200	0	90%	0	0	0%	
SPARE	480	1600	0	90%	0	0	0%	

FACTORY 12KV AND SUBSATION LOAD CALCULATION
NO SCALE

GENERATOR LOAD CALCULATION
NO SCALE

	RATINGS		CONN.		OPER.		%	
FEEDER ID	VOLTS	AMPS	KVA	DIVERSITY	KVA	AMPS	LOADING	NOTES
GENERATOR 3 (DATA CENTER)	480	3000	1718		1546	1860	62%	2500KVA RATED GENERATOR PLANT (LESS DERATING)

	RATINGS		CONN.		OPER.		%	
Bus ID	VOLTS	AMPS	KVA	DIVERSITY	KVA	AMPS	LOADING	NOTES
1FOFO-EHDP-2 04	480	1600	857		772	928	58%	
1FOFO-UPS-LDP-2 01A	480	800	525	90%	473	568	71%	75 RACKS @7000W EACH, 525KVA
1FOFO-EHDP-2 04A	480	800	332	90%	299	360	45%	

	RATINGS		CONN.		OPER.		%	
Bus ID	VOLTS	AMPS	KVA	DIVERSITY	KVA	AMPS	LOADING	NOTES
1FOFO-EHDP-2 05	480	1600	861		775	932	58%	
1FOFO-UPS-LDP-2 02A	480	800	525	90%	473	568	71%	75 RACKS @7000W EACH, 525KVA
1FOFO-EHDP-2 05A	480	800	336	90%	302	364	45%	

ELECTRICAL EQUIPMENT, SUCH AS SWITCHBOARDS, SWITCHGEAR, PANELBOARDS, INDUSTRIAL CONTROL PANELS, METER SOCKET ENCLOSURES IN OTHER THAN DWELLING UNITS SHALL BE FIELD OR FACTORY MARKED TO WARN QUALIFIED PERSONS OF POTENTIAL ELECTRIC ARC FLASH HAZARDS. NEC 110.16(A). IN ADDITION TO THE REQUIREMENTS OF 110.16(A), A PERMANENT LABEL SHALL BE FIELD OR FACTORY APPLIED TO SERVICE EQUIPMENT RATED 1200 AMPS OR MORE. THE LABELS SHALL MEET THE REQUIREMENTS OF NEC 110.21(B).